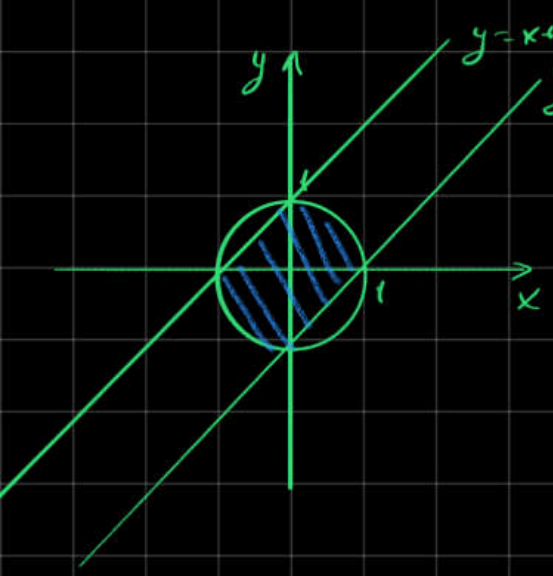
2 7 12 13 16 21 26 38 39 47 51 61 65 83 104 85 91 112 130

2) $f(x,y) = \arccos(y-x) + \ln(1-x^2-y^2)$ tanım kümesini göster.

$-1 \le \arccos x \le 1 \rightarrow -1 \le y-x \le 1$ $\cap$ $1-x^2-y^2 > 0$

$y = x-1$ $y = x+1$ $x^2+y^2 < 1$

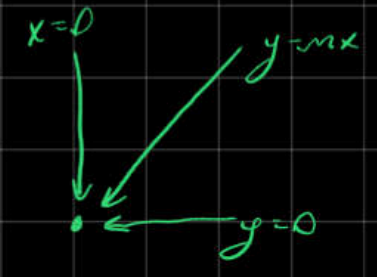


$TK = \{(x,y) \in \mathbb{R}^2 : y \le 1+x, y > x-1, x^2+y^2 < 1\}$

7) $\lim_{(x,y)\to(0,0)} \frac{5(x+2y^3)^2}{x^2+4y^6}$

a) $x=0$ doğrusu üzerinden yaklaşın.

$\lim_{y\to 0} \frac{5(2y^3)^2}{4y^6} = \lim_{y\to 0} 5 = 5$ ✓



b) $y=0$ " " "

$\lim_{x\to 0} \frac{5x^2}{x^2} = 5$ ✓

c) $y=mx$ " " "

$$\lim_{x\to 0}\frac{5(x+2(mx)^3)^2}{x^2+4(mx)^6}=\lim_{x\to 0}\frac{5(x+2m^3x^3)^2}{x^2+4m^6x^6}=\frac{5(x^2+4m^6x^6+4m^3x^4)}{x^2+4m^6x^4}$$

$$\frac{5(x^2(1+4m^6x^4+4m^3x^2)}{x^2(1+4m^6x^4)}$$

$$\frac{5(1+4m^6x^4)}{(1+4m^6x^4)}+\frac{4m^3x^2}{(1+4m^6x^4)}$$

$$5+0=5$$

12) $\lim_{(x,y)\to(0,0)}\frac{2x^2+2y^2}{\sqrt{x^2+y^2+9}-3}=?$

$$\lim_{(x,y)\to(0,0)}\frac{2x^2+2y^2}{\sqrt{x^2+y^2+9}-3}=\frac{0}{0}\ \text{bls}\rightarrow\lim_{(x,y)\to(0,0)}\frac{2x^2+2y^2(\sqrt{x^2+y^2+9}+3)}{(\sqrt{x^2+y^2+9}-3)(\sqrt{x^2+y^2+9}+3)}$$

$$=\lim_{(x,y)\to(0,0)}\frac{2(x^2+y^2)(\sqrt{x^2+y^2+9}+3)}{x^2+y^2}=12$$

13) $f(x,y)=\begin{cases}\frac{(x+y)\sin(x^2+y^2)}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$ fonks. sürekli midir?

$$\lim_{(x,y)\to(0,0)}\frac{(x+y)(\sin x^2+y^2)}{x^2+y^2}\overset{?}{=}f(0,0)=0$$

$$\lim_{(x,y)\to(0,0)} \frac{(x+y)(\sin x^2+y^2)}{x^2+y^2} = \frac{0}{0} \text{ bls.}$$

$$\begin{pmatrix} x = r\cos\theta \\ y = r\sin\theta \\ r^2 = x^2+y^2 \end{pmatrix}$$

$$\lim_{r\to 0} \frac{(r\cos\theta + r\sin\theta)(\sin r^2)}{r^2} = \lim_{r\to 0} r(\cos\theta+\sin\theta)\cdot\frac{(\sin(r^2))}{r^2}$$

$r(\cos\theta+\sin\theta) \to 0$; $\frac{(\sin(r^2))}{r^2} \to 1$; $0\cdot 1 = 0$

0 süreklidir

20) $w = (x+y)^{xy}$ için $\frac{\partial w}{\partial x}$ ve $\frac{\partial w}{\partial y}$ = ?

$$\ln(w) = \ln((x+y)^{xy}) = xy\ln(x+y)$$

$$w = e^{xy\ln(x+y)}$$

$$\frac{\partial w}{\partial x} = \underbrace{e^{xy\ln(x+y)}}_{(x+y)^{xy}} \cdot \underbrace{(xy\ln(x+y))'}_{y\ln(x+y)+\frac{xy}{x+y}} = (x+y)^{xy}\cdot\left(y\ln(x+y)+\frac{xy}{x+y}\right)$$

$$y\ln(x+y) + \underbrace{\frac{\partial \ln(x+y)}{\partial x}}_{\frac{1}{x+y}}\cdot(xy)\Big\}\; y\ln(x+y)+\frac{xy}{x+y}$$

(16) $f(x,y) = \begin{cases} \frac{2x^3+3y^3}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ ise $f_x(0,0)$ ve $f_y(0,0) = ?$

$$f_x(x_0,y_0) = \lim_{h\to 0} \frac{f(x_0+h, y_0) - f(x_0,y_0)}{h}$$

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{2h^3}{h^2} - 0}{h} = 2$$

$$f_y(x,y) = \lim_{h\to 0} \frac{f(x, y+h) - f(x,y)}{h}$$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{3h^3}{h^2} - 0}{h} = 3$$

(21) $\frac{\partial u}{\partial t} = c \cdot \frac{\partial^2 u}{\partial x^2}$ tipindeki denklere yayılma denklemi denir.

$u(x,t) = \frac{1}{\sqrt{t}} e^{-x^2/kt}$ fonk. hangi c sabiti için yayılma denk. sağlar?

$$u_t = \frac{-1}{2} \cdot t^{-3/2} \cdot e^{-x^2/kt} + \frac{1}{\sqrt{t}} \left(e^{\frac{-x^2}{kt}} \cdot \frac{x^2}{k} \cdot t^{-2} \right) = e^{\frac{-x^2}{kt}} \left(\frac{-1}{2} t^{-3/2} + \frac{x^2}{k} t^{-5/2} \right)$$

$$u_x = t^{-1/2} \cdot e^{-x^2/kt} \cdot \frac{-2x}{k} \cdot t^{-1} = e^{-x^2/kt} \cdot t^{-3/2} \cdot \frac{-2x}{k}$$

$$u_{xx} = e^{-x^2/kt} \cdot t^{-3/2} \cdot \frac{-2}{k} + e^{-x^2/kt} \cdot \frac{-2x}{kt} \cdot t^{-3/2} \cdot \frac{-2x}{k}$$

$$= e^{-x^2/4t}\left(t^{-3/2}\cdot\frac{-2}{2} + \frac{4x^2}{4^2}\cdot t^{-5/2}\right)$$

$$e^{\frac{-x^2}{4t}}\left(\frac{-1}{2}t^{-3/2} + \frac{x^2}{4}t^{-5/2}\right) = \frac{c}{d} \cdot e^{-x^2/4t}\left(t^{-3/2}\cdot\frac{-2}{2} + \frac{4x^2}{4^2}\cdot t^{-5/2}\right)$$

$c/d = 1/4$

26) $w = xy + z^2$, $x = u^2e^v$, $y = u\cos v$, $z = v\sin u$ old. göre $\frac{\partial w}{\partial u}$?

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial u}$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$y \quad 2u\cdot e^v \quad x \quad \cos v \quad 2z \quad v\cdot\cos u$$

$$= y\cdot 2u\cdot e^v + x\cdot\cos v + 2z\, v\cdot\cos u$$

$v = 0$, $x = \pi^2$
$u = \pi$, $y = \pi$
$z = 0$

$\Big\}$ $\pi\cdot 2\pi + \pi^2 = 3\pi^2$

38) $\frac{\partial^2 z}{\partial t^2} = a^2\frac{\partial^2 z}{\partial x^2}$, $(a \neq 0)$ bir boyutlu dalga denklemde $u = x - at$, $v = x + at$ dönş. yapılırsa, denklem alacağı yeni?

$$z_{tt} = a^2\cdot z_{xx} \qquad \begin{pmatrix} u = x - at \\ v = x + at \end{pmatrix}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$\downarrow \qquad \downarrow$$

$$1 \qquad 1$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v}\right)\left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\right) = \frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v \partial u} + \frac{\partial^2 z}{\partial v^2}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u} \cdot \underbrace{\frac{\partial u}{\partial t}}_{-a} + \frac{\partial z}{\partial v} \cdot \underbrace{\frac{\partial v}{\partial t}}_{a}$$

$$\frac{\partial z}{\partial t} = a\left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\right)$$

$$\frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial z}{\partial t}\right) = a\left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v}\right)\left(a\left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}\right)\right) = a^2\left(\frac{\partial^2 z}{\partial u^2} - 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}\right)$$

$$\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

$$a^2\left(\frac{\partial^2 z}{\partial u^2} - 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}\right) = a^2\left(\frac{\partial^2 z}{\partial u^2} + 2\frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}\right)$$

$$\rightarrow \frac{\partial^2 z}{\partial u \partial v} = 0$$

39) $f(x) = \int_0^x \ln(\arcsin y)\cos(x-y)\,dy$ ise, $f''(x)+f(x) = ?$

$f'(x) = \int_0^x \ln(\arcsin y)\cdot -\sin(x-y)\,dy + \ln(\arcsin x)\cdot\cos 0 \cdot 1 - 0$

$= \ln(\arcsin x) - \int_0^x \ln(\arcsin y)\cdot\sin(x-y)\,dy$

$f''(x) = \dfrac{\frac{1}{\sqrt{1-x^2}}}{\arcsin x} - \int_0^x \ln(\arcsin y)\cos(x-y)\,dy + \ln(\arcsin 0)\cos 0\cdot 1 - 0$

$f''(x) = \dfrac{1}{\sqrt{1-x^2}\arcsin x} - \int_0^x \ln(\arcsin y)\cos(x-y)\,dy$

$\dfrac{1}{\sqrt{1-x^2}\arcsin x} - \int_0^x \ln(\arcsin y)\cos(x-y)\,dy + \int_0^x \ln(\arcsin y)\cos(x-y)\,dy$

47)

(46) $z^3 + xyz = 1 - xy^2$ denklemi ile tanımlı $z = z(x,y)$ fonk. için $z_x(1,1)$, $z_y(1,1)$ kısmi türevlerini bulunuz.

$$z_x = \frac{\partial z}{\partial x} = 3z^2 \cdot \frac{\partial z}{\partial x} + y\left(1 \cdot z + \frac{\partial z}{\partial x} \cdot x\right) = -y^2$$

Δ (y sabit, z x'e bağlı fonk., x'e göre türev)

$$= 3z^2 \cdot \frac{\partial z}{\partial x} + yz + \frac{\partial z}{\partial x} xy + y^2 = 0$$

$$\left(\frac{\partial z}{\partial x}\right)(3z^2 + xy) = -y^2 - yz$$

$$\frac{\partial z}{\partial x} = \frac{-y^2 - yz}{3z^2 + xy}$$

$$\frac{\partial z}{\partial x}(1,1,0) = \frac{-1}{1} = -1$$

($(1,1,z)$: $z^3 + z = 0$, $z(x^2+1) = 0$, $z = 0$ veya $x^2 = -1$ reel değil)

51) $\sqrt{(5,98)^2 + (8,01)^2}$ sayısının yaklaşık değeri?

$\Delta_z = f_x\,dx + f_y\,dy$

- $f(x,y) = \sqrt{x^2+y^2}$
- hedef: $(5,98;\ 8,01)$ — $dx = -0,02$
- merkez: $(6,8)$ — $dy = +0,01$
- $f(6,8) = \sqrt{36+64} = 10$
- $f_x = \frac{x}{\sqrt{x^2+y^2}}$ $f_x(6,8) = 0,6$ / $f_y = \frac{y}{\sqrt{x^2+y^2}}$ $f_y(6,8) = 0,8$

$df = f_x(6,8)\,dx + f_y(6,8)\,dy = -0,004$

$f(5,98;\ 8,01) \approx f(6,8) + df = 10 - 0,004 = 9,996$

6) $f(x,y) = x^3 - 3xy + y^3$ fonksiyonunun kritik noktalarının cinsi?

$f_x = 3x^2 - 3y = 0 \quad 3y^4 - 3y = 0$

$f_y = -3x + 3y^2 = 0 \quad y(y^3 - 1) = 0$

$x = y^2 \qquad y = 0 \vee y = 1$

$x = 0 \vee x = 1$

$(0,0)$ ve $(1,1)$

$D = f_{xx} \cdot f_{yy} - (f_{xy})^2$

$f_{xx} = 6x \quad f_{xy} = -3$

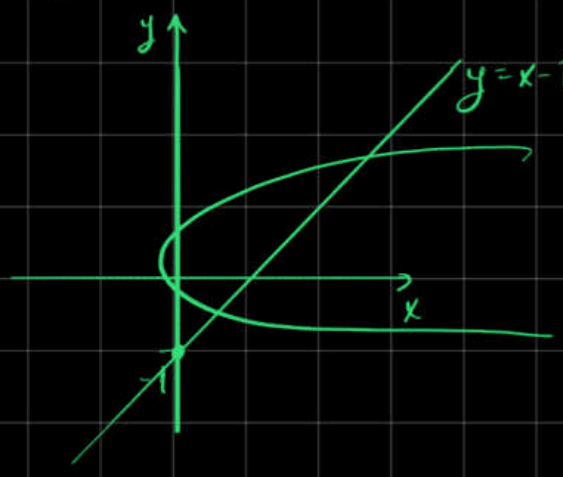
$f_{yy} = 6y$

$D = 36xy - 9$

$(0,0)$ için $D = -9 \leq 0$ → eyer noktası

$(1,1)$ için → $D = 27 \geq 0$ ve $f_{xx} = 6 \geq 0$ → min noktası

65) $f(x,y) = 2x^2 - 3y^2 + 2x + 3$ fonks. $y^2 = 2x+3$ parabolü, $y = x-1$ doğrusu arasında kalan kapalı B bölgesi üzerindeki mutlak ekstremumlarını bulunuz.



$x = y+1 \rightarrow y^2 = 2(y+1)+3$

$y^2 = 2y + 5$

$y^2 - 2y - 5 = 0$

$y_1 = 1+\sqrt{6} \quad \vee \quad 1-\sqrt{6} = y_2$ aday

$x_1 = 2+\sqrt{6} \quad 2-\sqrt{6} = x_2$

$f_x = 4x + 2$

$f_y = -6y \quad y = 0 \quad x = -\frac{1}{2} \quad (-\frac{1}{2}, 0)$ aday

$g(x) = 2x^2 - 3(2x+3) + 2x + 3 \rightarrow g'(x) = 4x - 4 = 0$

$x = 1 \rightarrow y_{1,2} = \mp\sqrt{5}$

aday: $(1, \sqrt{5})$

$h(x) = 2x^2 - 3(x-1)^2 + 2x + 3 \rightarrow h'(x) = -2x + 8 = 0$

$x_1 = 4 \quad y_1 = 3$

aday $(4,3)$

$f(-\frac{1}{2}, 0) = 2,5$

$f(1, \sqrt{5}) = -8 \rightarrow$ mm mutlak

$f(4,3) = 16 \rightarrow$ max mutlak

83) Yüzleri koordinat düzlemlerine paralel olacak şekilde $16x^2+4y^2+9z^2=144$ elipsoidi içine dikdörtgen prizması yerleştirilecektir. Bu prizmanın hacmi max?

$g(x,y,z) = 16x^2+4y^2+9z^2-144$, $V(x,y,z) = 2x\,2y\,2z = 8xyz$

$\nabla V = \lambda \nabla g$

$f_x = \lambda g_x$ $\quad f_y = \lambda g_y$ $\quad f_z = \lambda g_z$

$8yz = \lambda . 32x$ $\quad 8xz = \lambda . 8y$ $\quad 8xy = \lambda 18z$

$32x^2 = 8y^2 = 18z^2$

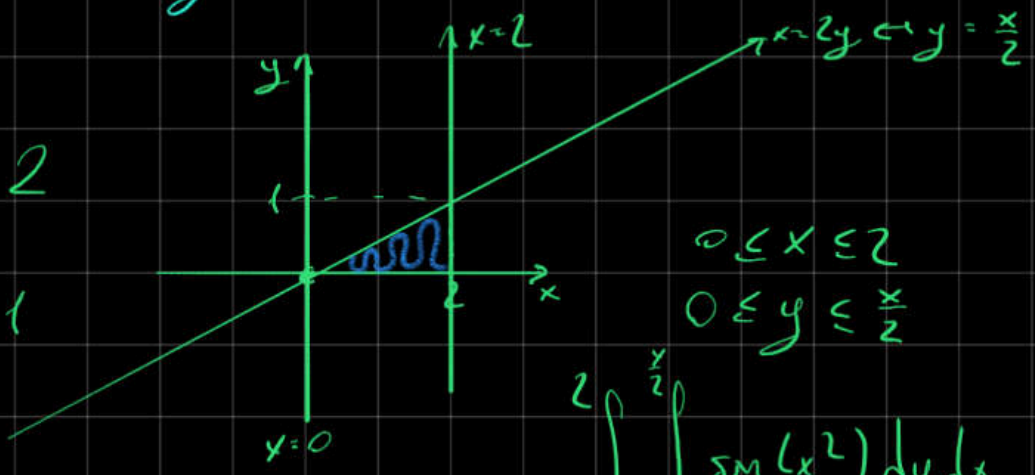
$y^2 = 4x^2$, $z^2 = \frac{16}{9}x^2$

$16x^2 + 4.4x^2 + 9.\frac{16}{9}x^2 = 144$

$x = \sqrt{3}$, $y = 2\sqrt{3}$, $z = 4\sqrt{3}$

$V_{max} = 8xyz = 8\sqrt{3}.2\sqrt{3}.\frac{4}{\sqrt{3}} = 64\sqrt{3}$

(Ödev) $\int_0^1 \int_{2y}^{2} \sin(x^2)\,dx\,dy$ int. için



Fonk. s. / $2y \le x \le 2$

Sayı s. / $0 \le y \le 1$

$0 \le x \le 2$

$0 \le y \le \frac{x}{2}$

$$\int_0^2 \int_0^{\frac{x}{2}} \sin(x^2)\,dy\,dx$$

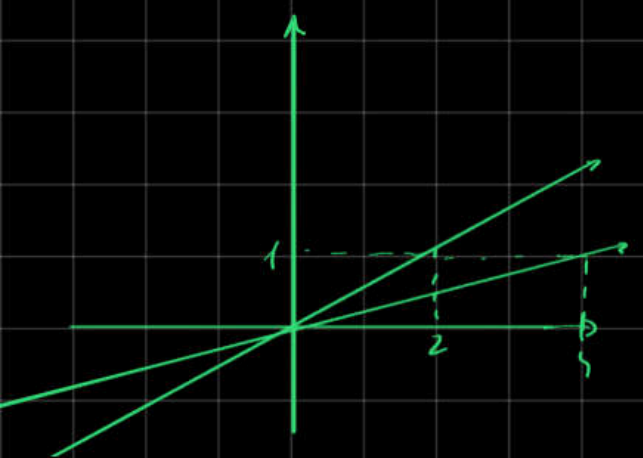
$$= \int_0^2 \sin(x^2).y = \frac{\sin(x^2).x}{2}$$

$$= \int_0^2 \frac{x}{2}\sin(x^2) = \frac{1}{4}\int_0^4 \sin(u)\,du$$

$$\left(\begin{array}{l} u = x^2 \\ du = 2x\,dx \end{array}\right)$$

$$= \frac{1}{4}\left(-\cos(u)\right)\Big|_0^4 = \frac{1-\cos 4}{4}$$

8S) R bölgesi 1. bölgedeki $x^2-y^2=1$, $x^2-y^2=9$, $xy=2$, $xy=4$ hiperbolleri sınırlanan bölge olduğuna göre $\iint_R (x^2+y^2)\,dA$ int. ?



$1 \le x^2-y^2 \le 9$ ⟷ $1 \le u \le 9$

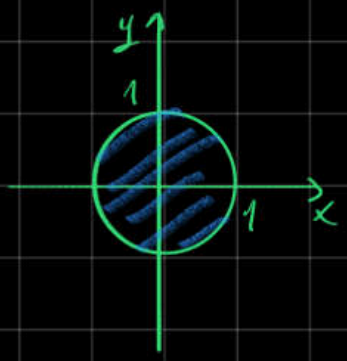
$2 \le xy \le 4$ ⟷ $2 \le v \le 4$

$u_x = 2x$, $u_y = -2y$, $v_x = y$, $v_y = x$ $\quad |J| = \begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix} = 2x^2 + 2y^2$

$\iint f(x^2-y^2, xy)(2x^2+2y^2)$

SG AMK

11) B bölgesi $x^2+y^2 \le 1$ dairesi old. göre $\iint_B (1-x^2-y^2)\,dxdy$?



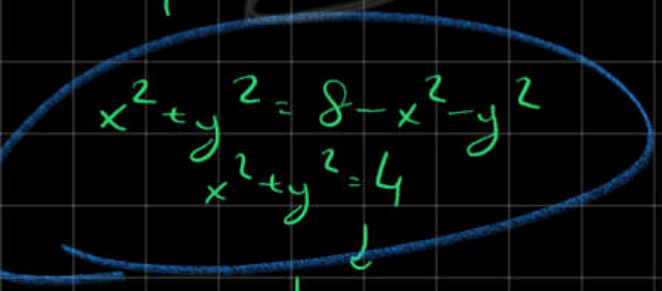
$0 \le r \le 1$
$0 \le \theta \le 2\pi$

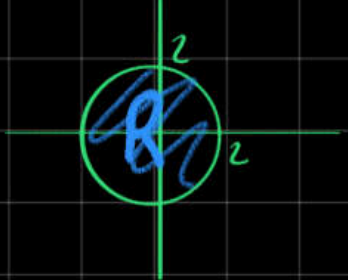
$$\rightarrow \int_0^{2\pi}\int_0^1 (1-r^2)\,r\,dr\,d\theta$$

$$\int_0^1 (r-r^3)\,dr = \left(\frac{r^2}{2} - \frac{r^4}{4}\right)\Big|_0^1 = \frac{1}{4}$$

$$\frac{1}{4}\theta\Big|_0^{2\pi} = \frac{\pi}{2}$$

12) Alt yüzey: $z = x^2+y^2$ paraboloidi ve üst yüzey, $z = 8-x^2-y^2$ par. oldu cismin hacmi?





$0 \le r \le 2$
$0 \le \theta \le 2\pi$

$z_{üst} - z_{alt} = (8-x^2-y^2-x^2-y^2)$

$= 8 - 2(x^2+y^2)$

$= 8 - 2r^2$

$$V = \iint_R (z_{üst} - z_{alt})\,dA$$

$$= \int_0^{2\pi} \int_0^{2} (8 - 2r^2) \, r \, dr \, d\theta = 16\pi$$

130) $x^2 + y^2 + z^2 = 25$ küresinin $z = 3$, $z = 4$ düzlemleri arasında kalan kısmın alanı?

Küre yarıçap = 5

$h = 4 - 3 = 1$

$A = 2\pi . r . h$

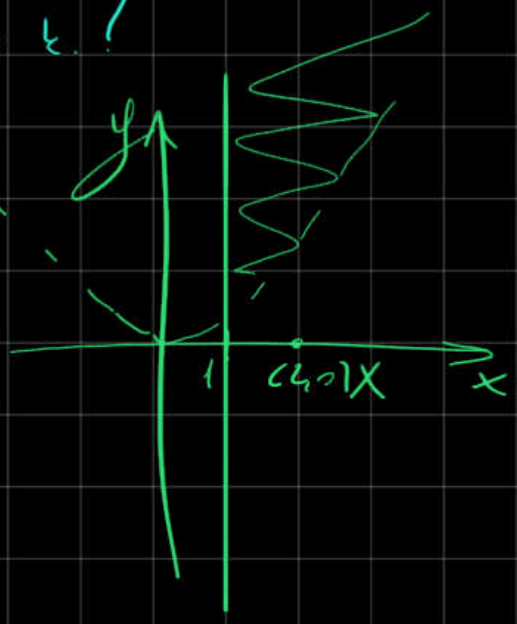
$= 2\pi . 5 . 1 = 10\pi$

4) $f(x,y) = \sqrt{y-x^2} + \ln(x-1)$ tanım k.?

$y - x^2 \geq 0$ $\quad x - 1 > 0$

$y \geq x^2$ $\quad x > 1$

$\{(x,y) \in \mathbb{R}^2 \mid y \geq x^2,\ x > 1\}$



48) $f(x,y) = \ln(x + \sqrt{x^2+y^2})$ fonk. için $f_x(1,0) + f_y(0,1) = ?$

$f_x = \dfrac{\left(1 + \dfrac{1}{\sqrt{x^2+y^2}}\right)}{x + \sqrt{x^2+y^2}}$

$(x^2+y^2)^{1/2} \rightarrow \frac{1}{2}(x^2+y^2)^{-1/2}(2x)$

$f_y = \dfrac{\dfrac{1}{\partial x^2+y^2}}{x + \sqrt{x^2+y^2}}$

$f_x(1,0) = \dfrac{1 + \frac{1}{1}}{2} = 1$

$f_y(0,1) = \dfrac{\frac{1}{1}}{1} = 1$

$+ \Big) 2$

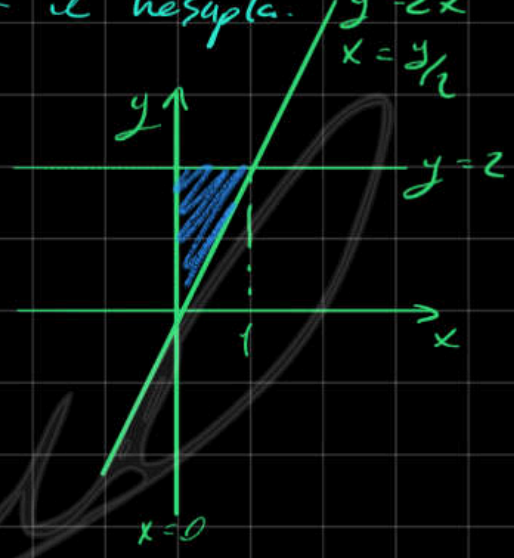
44) $yz - \ln z = x + y$ ise $\dfrac{\partial z}{\partial x} = ?$

$y\dfrac{\partial z}{\partial x} - \left(\dfrac{\frac{\partial z}{\partial x}}{z}\right) = 1$ $\quad \begin{pmatrix} y \text{ sabit} \\ z \ x'\text{e bağlı fonk.} \\ x'\text{e göre türev} \end{pmatrix}$

$$yz\frac{\partial z}{\partial x} - \frac{\partial z}{\partial x} = z$$

$$\frac{\partial z}{\partial x}(yz-1) = z \rightarrow \frac{\partial z}{\partial x} = \frac{z}{yz-1}$$

(402) $\int_0^1 \int_{2x}^{2} e^{y^2}\,dy\,dx$ mt sırasını değiştir ve hesapla.



sayı s.: $0 \le x \le 1$

fonk. s.: $2x \le y \le 2 \rightarrow y=2x,\ y=2$

sayı s: $0 \le y \le 2$

fonk. s.: $0 \le x \le y/2$

$$\int_0^2 \int_0^{y/2} e^{y^2}\,dx\,dy$$

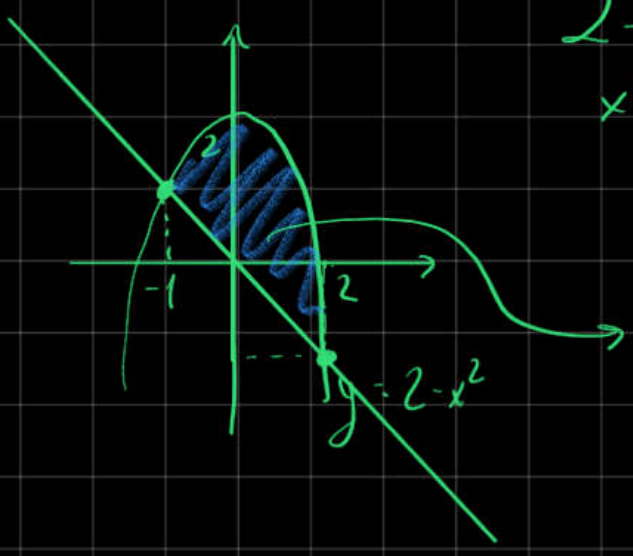
$$(e^{y^2}\cdot x)\Big|_0^{y/2} = e^{y^2}\cdot\frac{y}{2}$$

$$\int_0^2 e^{y^2}\frac{y}{2}\,dy = \frac{1}{4}\int_0^4 e^u\,du$$

$$\begin{pmatrix} y^2 : u \\ 2y\,dy : du \\ y/2\,dy : \frac{du}{4} \end{pmatrix}$$

$$= \frac{1}{4}\cdot e^{u}\Big|_0^4 = \frac{1(e^4-1)}{4}$$

407) $y = 2 - x^2$ parabolü ile $y = -x$ arasında kalan bölgenin alanı?



$$2 - x^2 = -x$$
$$x^2 - x - 2 = 0$$
$$-2 \cdot 1 \rightarrow x_1 = 2$$
$$1 \cdot -1 \rightarrow x_2 = -1$$

sayı sınırı: $-1 \le x \le 2$

f. s.: $-x \le y \le 2 - x^2$

$$\int_{-1}^{2} \int_{-x}^{2-x^2} 1 \, dy \, dx$$

$$y \Big|_{-x}^{2-x^2} = 2 - x^2 + x$$

$$\int_{-1}^{2} (-x^2 + x + 2) \, dx = \left(\frac{-x^3}{3} + \frac{x^2}{2} + 2x \right) \Big|_{-1}^{2}$$

$$\left(\frac{-8}{3} + 2 + 4 \right) - \left(\frac{1}{3} + \frac{1}{2} - 2 \right)$$

$$8 - 3 - \frac{1}{2} = 5 - \frac{1}{2} = \frac{9}{2}$$

68) $(x-2)^2+(y+3)^2=1$ için $x+y$ nin max değeri?

$f(x,y)=x+y$ nin max?

$$g(x,y)=(x-2)^2+(y+3)^2-1=x^2-4x+4+y^2+6y+9-1$$
$$=x^2+y^2-4x+6y+12$$

$$F_x=\lambda\cdot g_x \Rightarrow 1=\lambda\cdot(2x-4) \rightarrow 1=2x\lambda-4\lambda \rightarrow x=\frac{1}{2\lambda}+2$$

$$F_y=\lambda\cdot g_y \Rightarrow 1=\lambda\cdot(2y+6) \rightarrow 1=2y\lambda+6\lambda \rightarrow 2y\lambda=\frac{1}{2\lambda}-3$$

$$\frac{1}{4\lambda^2}+4+\frac{2}{\lambda}+\frac{1}{4\lambda^2}+9-\frac{3}{\lambda}-4\left(\frac{1}{2\lambda}+2\right)+6\left(\frac{1}{2\lambda}-3\right)+12=0$$

$$-\frac{2}{\lambda}-8+\frac{3}{\lambda}-18+12$$

$$\frac{1}{2\lambda^2}+4+9-8-6$$

$$\frac{1}{2\lambda^2}=1 \qquad \lambda^2=\frac{1}{2}$$

$$\boxed{\lambda_1=\frac{1}{\sqrt{2}} \quad \lambda_2=\frac{-1}{\sqrt{2}}}$$

$\lambda_1=\frac{1}{\sqrt{2}}$ için $x=\frac{1}{2\cdot\frac{1}{\sqrt{2}}}+2=\frac{\sqrt{2}}{2}+2$, $y=\frac{\sqrt{2}}{2}-3$

$\lambda_2=\frac{-1}{\sqrt{2}}$ için $x=\frac{-\sqrt{2}}{2}+2$, $y=\frac{-\sqrt{2}}{2}-3$

$f\left(\frac{\sqrt{2}}{2}+2,\frac{\sqrt{2}}{2}-3\right)=\sqrt{2}-1\}$ max

$f\left(\frac{-\sqrt{2}}{2}+2,\frac{-\sqrt{2}}{2}-3\right)=-\sqrt{2}-1\}$ min

(67) $f(x,y) = 2x^2 - 3y^2 + 2x + 3$ fonksiyonunun $B = \{(x,y) \mid (x,y) \in \mathbb{R}, x^2 + y^2 \leq 4\}$ bölgesi üzerindeki ekstrem değerini araştırınız.

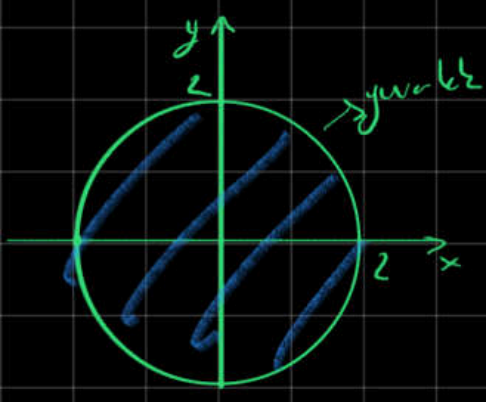
$f_x(x,y) = 4x + 2 = 0 \quad x = -\frac{1}{2}$

$f_y(x,y) = -6y = 0 \quad y = 0$

$\Rightarrow (-\frac{1}{2}, 0)$ içerdeyse kritik aday → yerel noktası

$f_{xx} = 4 \quad f_{xy} = 0 \quad D = 4 \cdot -6 \cdot 0^2 = -24 < 0$

$f_{yy} = -6$



$g(x,y) = x^2 + y^2 - 4 \quad g_x = 2x, \; g_y = 2y$

$f_x = \lambda \cdot g_x \rightarrow 4x + 2 = \lambda \cdot 2x \rightarrow x = \frac{1}{\lambda - 2}$

$f_y = \lambda \cdot g_y \rightarrow -6y = \lambda \cdot 2y \rightarrow 2y(\lambda + 3) = 0$

$y = 0 \quad \lambda + 3 = 0 \rightarrow \lambda_2 = -3$

$4x + 2 = \lambda \cdot 2x$

$\frac{4x+2}{2x} = \lambda$

$2 + \frac{1}{x} = \lambda$

$\lambda - 2 = \frac{1}{x}$

$\frac{1}{\lambda - 2} = x$

$\left(\frac{1}{\lambda - 2}\right)^2 + 0 = 4$

$\frac{1}{\lambda^2 - 4\lambda + 4} = 4 \quad 1 = 4\lambda^2 - 16\lambda + 16$

$4\lambda^2 - 16\lambda + 15 = 0$

$2 \quad -3 \quad (2\lambda - 3)(2\lambda - 5) = 0$

$2 \quad -5 \quad \lambda \quad \frac{3}{2} \quad \lambda \quad \frac{5}{2}$

$\lambda_1 = \frac{3}{2}$ için $x = -2, \; y = 0 \quad (-2, 0)$

$\lambda_2 = \frac{5}{2}$ için $x = 2, \; y = 0 \quad (2, 0)$

$f(-2,0) = 7$ yerel max

$f(2,0) = 15$ mutlak max

$d_y = -3$ için $x = \frac{1}{-5}$ $\quad x^2 + y^2 = 4$ $\quad \rightarrow \left(-\frac{1}{5}, \frac{\sqrt{99}}{5}\right), \left(-\frac{1}{5}, -\frac{\sqrt{99}}{5}\right)$

$\frac{1}{25} + y^2 = \frac{100}{25}$

$y^2 = \frac{99}{25}$ $\quad y_1 = \frac{\sqrt{99}}{5}$ $\quad y_2 = -\frac{\sqrt{99}}{5}$

$f\left(-\frac{1}{5}, \frac{\sqrt{99}}{5}\right) = -\frac{230}{25}$

$f\left(-\frac{1}{5}, -\frac{\sqrt{99}}{5}\right) = -\frac{270}{25}$

} mutlak min

93) $\int_1^2 \int_{\frac{1}{y}}^{y} \sqrt{\frac{y}{x}}\, e^{\sqrt{xy}}\, dx\, dy$ alt. $x = u + \frac{v}{2}$, $y = v$ dönüşür.

$1 \le y \le 2 \rightarrow y = 1,\ y = 2,\ 1 \le v \le 2$

$\frac{1}{y} \le x \le y \rightarrow x = \frac{1}{y},\ x = y,\ \frac{1}{v} \le u + \frac{v}{2} \le v$

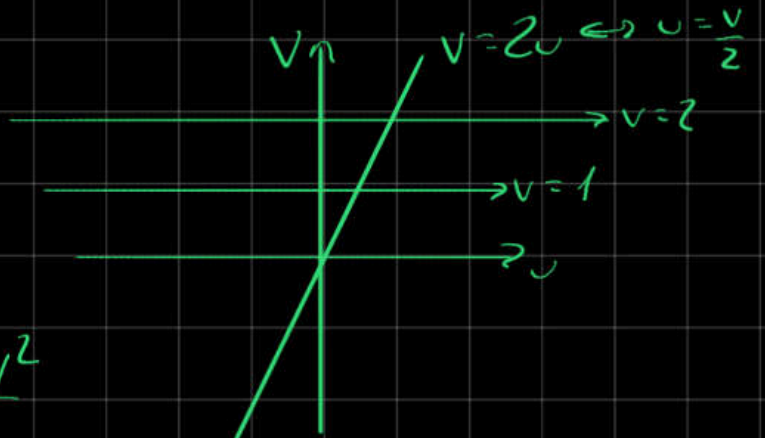
$\frac{2 - v^2}{2v} \le u \le \frac{v}{2}$

$y = 1 \rightarrow v = 1$

$y = 2 \rightarrow v = 2$

$x = \frac{1}{y} \rightarrow u + \frac{v}{2} = \frac{1}{v} \rightarrow u = \frac{2 - v^2}{2v}$

$x = y \rightarrow 2u = v$



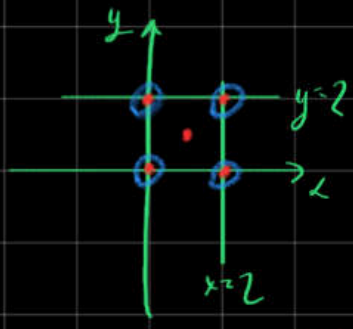
$$|J| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1$$

$$\int_1^2 \int_{\frac{2-v^2}{2v}}^{\frac{v}{2}} f\left(\frac{u+v}{2}, v\right) du\,dv = \int_1^2 \int_{\frac{2-v^2}{2v}}^{\frac{v}{2}} \sqrt{\frac{v}{\frac{u+v}{2}}} \cdot e^{\sqrt{\frac{u+v}{2} \cdot v}}\, du\,dv$$

791)

α $f(x,y) = 2x + 2y - x^2 - y^2$ fonksiyonunun $x=0$, $x=2$, $y=0$, $y=2$ doğruları tarafından sınırlanan bölgedeki mutlak maksimum ve minimum değerlerini bulunuz.

$f_x = 2 - 2x = 0 \quad x = 1 \quad (1,1)$ aday
$f_y = 2 - 2y = 0 \quad y = 1$



$f(x,y)$ ⟵ ? türev

$y=2 \rightarrow 2x - x^2 \longrightarrow 2 - 2x = 0 \quad x = 1 \quad (1,2)$

$y=0 \rightarrow 2x - x^2 \rightarrow 2 - 2x = 0 \quad x = 1 \quad (1,0)$

$x=0 \rightarrow 2y - y^2 \rightarrow 2 - 2y = 0 \quad y = 1 \quad (0,1)$

$x=2 \rightarrow 2y - y^2 \rightarrow 2 - 2y = 0 \quad y = 1 \quad (2,1)$

1) $\lim_{(x,y)\to(0,0)} \frac{5(x+2y^3)^2}{x^2+4y^6}$

a) $y=x^2$ eğrisinden yaklaş

$$\lim_{x\to 0} \frac{5(x+2x^6)^2}{x^2+4x^{12}} = \frac{5(x^2+4x^7+4x^{12})}{x^2+4x^{12}} = \frac{5\cancel{x^2}(1+4x^5+4x^{10})}{\cancel{x^2}(1+4x^{10})}$$

$$= 5$$

b) $x=y^3$ eğrisi için

$$\lim_{y\to 0} \frac{5(3y^3)^2}{5y^6} = \frac{5\cdot 9y^6}{5y^6} = 9$$

$(0,0)$'da limit yok

2) $w=f(x,y)=x^2e^y$, $x=t^2+s^2$, $y=t^2s$, $(t,s)=(0,1)$

$\frac{\partial w}{\partial t}, \frac{\partial w}{\partial s}$?

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t}$$

$$= 2x\cdot e^y \cdot 2t + x^2e^y \cdot 2ts \longrightarrow 0+0=0$$

$(t,s)=(0,1)$ için $x=1$, $y=0$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s}$$

sin, cos $u, a^x, \ln_y$ türev olmaya

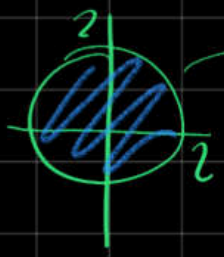
TAMAM

$= 2x \cdot e^y \cdot 2s + e^y \cdot x^2 \cdot t^2$

$= 2 \cdot 1 \cdot 2 = 4$

3) $z = 4 - x^2 - y^2$ paraboloidi ile $z = 0$ düzlemi arasında hacim?

$x^2 + y^2 = 4$ $\quad$ $4 - r^2$



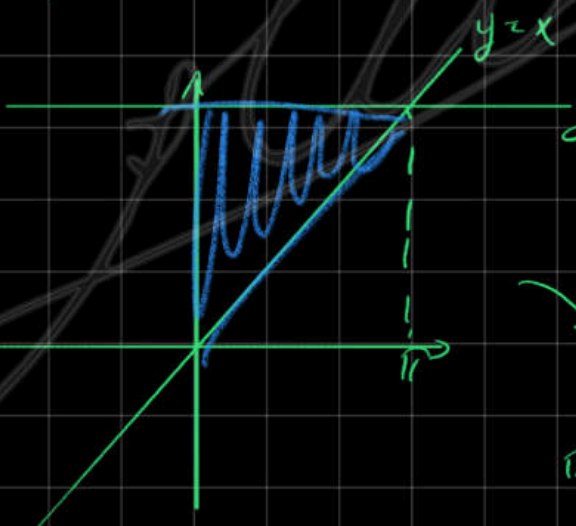
$\longrightarrow 0 \le r \le 2$, $0 \le \theta \le 2\pi$

$$\int_0^{2\pi} \int_0^2 (4 - r^2)\, r\, dr\, d\theta = \curvearrowright$$

4) $\int_0^{\pi} \int_x^{\pi} \frac{\sin y}{y}\, dy\, dx$?

$0 \le x \le \pi$

$x \le y \le \pi$ $\longrightarrow$



$\longrightarrow 0 \le x \le y$

$0 \le y \le \pi$

$$\int_0^{\pi} \int_0^{y} \frac{\sin y}{y}\, dx\, dy$$

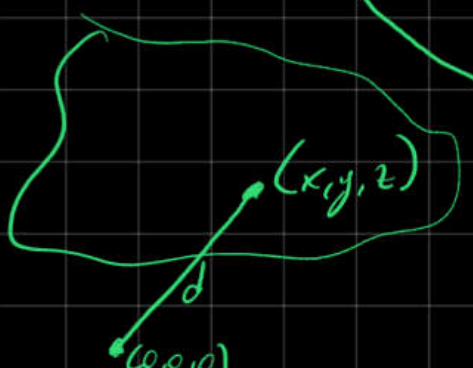
$$\left(\frac{\sin y}{y} \cdot x \right) \Big|_0^y$$

$$\int_0^{\pi} \sin y\, dy = (-\cos y) \Big|_0^{\pi}$$

$-\cos \pi - (-\cos 0)$

$-(-1) - (-1) = 2$

5) $z^2 = xy - 3x + 9$ yüzeyi ile orijine en yakın nokta?



$d = \sqrt{x^2+y^2+z^2}$

$f(x,y) = x^2+y^2+xy-3x+9$

$f_x = 2x+y-3 = 0 \quad -3y = 3 \quad y = -1$

$x = 2$

$f_y = 2y+x = 0 \rightarrow x = -2y \quad (2,-1)$

$D = f_{xx} \cdot f_{yy} - (f_{xy})^2 \qquad f_{xx} = 2 \quad f_{yy} = 2 \quad f_{xy} = 1$

$D = 2 \cdot 2 - 1^2 = 3 > 0$ ve $f_{xx} = 2 > 0$ old. $(2,-1)$ min

$z^2 = xy - 3x + 9 \longrightarrow (2,-1)$ için

$z^2 = -2 - 6 + 9 = 1$

$z_1 = 1 \vee z_2 = -1$

en yakın noktalar $= (2,-1,1)$

$(2,-1,-1)$